
When does overlap help? Diagnosing and fixing ASM/sASM for the cardiac solver stack

Overlap anomaly $\rightarrow$ spectral attribution $\rightarrow$ a non-GenEO coarse space $\rightarrow$ cross-system reuse (Sys1 $\rightarrow$ Sys2 $\rightarrow$ Sys3)

Sys1 monodomain
($1/\Delta t$) $M + \theta K$, Neumann, easy

Sys2 u_e recovery
pure-Neumann K , singular, hardest

Sys3 torso
different mesh, interface-Dirichlet

CG + PCASM(ICC) + two-level · exact-spectrum NumPy lab + PETSc 3.24

Roadmap

- I **The anomaly** — on SPD elliptic problems, unscaled BASIC ASM + inexact ICC + CG makes CG iterations *rise* with overlap (cases a–d, unstructured P1 FEM).
- II **Attribution** — which factor (dimension / boundary / shift) makes it hard, and through which spectral channel.
- III **The fix** — a non-GenEO two-level coarse: shift-invariant (K, M) spectral modes + known-coefficient A-harmonic modes.
- IV **Cross-system** — one build-once coarse reused across Sys1 → Sys2 (same mesh) and Sys3 (different mesh).

The unifying lens (all parts):

$$\text{iters} \approx \frac{1}{2} \sqrt{\kappa} \ln \frac{2}{\varepsilon}, \quad \kappa = \frac{\lambda_{\max}}{\lambda_{\min}}$$

$$\lambda_{\max}(\text{ASM}) \approx \omega \cdot \hat{N}$$

$$\lambda_{\max}(\text{sASM}) \approx \omega$$

$$\lambda_{\min} \sim 1/C_0^2, \quad C_0^2 \sim (1+H/\delta)$$

$\hat{N}$ =over-count (geometry), ω =inexact-ICC,
 $\lambda_{\min}$ =info transfer / coarse mode.

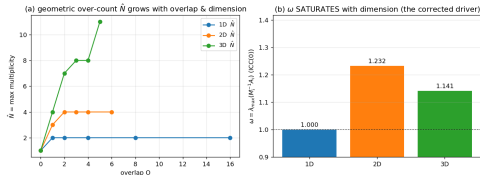
The mechanism: two spectral channels

ASM (BASIC): $M^{-1} = \sum_i R_i^\top A_i^{-1} R_i$ — overlap region added repeatedly $\Rightarrow$ over-count $\hat{N} = m_{\text{axis}}^d$.

sASM (weighted): $M^{-1} = D^{-1/2} (\sum_i R_i^\top A_i^{-1} R_i) D^{-1/2}$,
 $D = \text{diag}(m_k)$ — removes $\hat{N}$, stays symmetric (CG-safe).

ω **channel:** inexact ICC; $\omega=1$ in 1D (exact), saturates ~ 1.1 – 1.4 in 2D/3D.

$\lambda_{\min}$ **channel:** stable decomposition; collapses with weak boundary / small shift; the constant mode of pure-Neumann.



$\hat{N}$ grows $\propto m_{\text{axis}}^d$ with overlap & dimension; ω saturates.

Conclusion. sASM removes the $\hat{N}$ over-count ($\lambda_{\max}$ side); the remaining difficulty lives in $\lambda_{\min}$, which only a coarse space can fix.

Methods & tooling

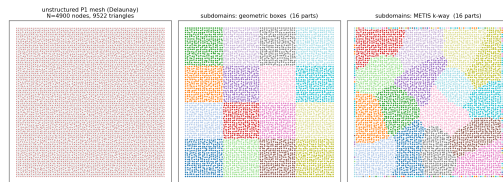
Operator: $-\nabla \cdot (A \nabla u) = f$ on $[0, 1]^d$; for the cardiac proxy $A = K + \sigma I$.

Discretisation: P1 unstructured (Delaunay) and structured FD; harmonic face coefficients.

Solver: CG + PCASM, ICC(0) (or exact) local solves; ASM vs sASM; optional two-level coarse.

Measured: CG iters, wall time, exact $\lambda_{\min}$, $\lambda_{\max}$, κ (similarity transform), $\hat{N}$, ω , CG-Ritz spectrum.

Platforms: NumPy/SciPy exact-spectrum lab (auditable) + PETSc 3.24 (scales & cross-validates).



Unstructured P1 mesh and the two subdomain partitions (geometric boxes / METIS).

Part I

The overlap anomaly

unstructured P1 FEM, cases a–d

Cases a–d: definition & parameters

Studies: does increasing ASM overlap reduce (classical) or *increase* CG iterations — and does sASM fix it?

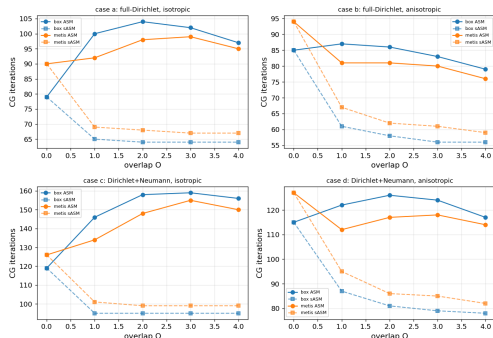
Case	Coefficient A	Boundary condition	Problem size
a	isotropic $A=I$	full Dirichlet	1D $N=4000$ / 2D $N=4900$ / 3D $N=8000$
b	anisotropic 2D $\text{diag}(10,1)$; 3D Niederer $\sigma_L/\sigma_T=7.58$ ($\parallel x$)	full Dirichlet	2D $N=4900$ (9522 tri) / 3D $N=8000$ (49k tet)
c	isotropic $A=I$	1 Dirichlet face + Neumann	1D/2D/3D as a
d	anisotropic (as b)	mixed (as c)	2D/3D as b

Parameters (all cells): $n_{\text{sub}} = 8(1\text{D})/16(2\text{D})/27(3\text{D})$; partition $\in \{\text{box}, \text{METIS}\}$; overlap $O \in \{0, 1, 2, 3, 4\}$; local ICC(0); CG rtol 10^{-8} ; scheme $\in \{\text{ASM=BASIC}, \text{sASM}\}$.

Conclusion. Clean baseline (a) isolates the anomaly; b/c/d add anisotropy and weak boundary to expose how it scales.

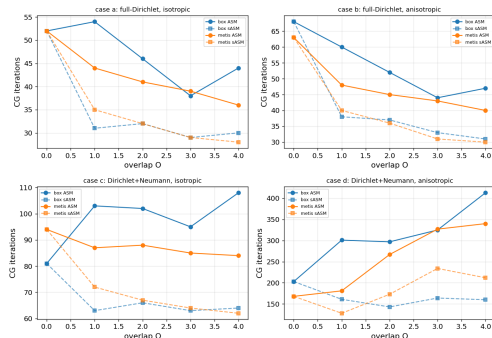
Result: ASM vs sASM iterations (2D & 3D)

2D unstructured P1 FEM: ASM vs sASM (CG iterations)



2D

3D unstructured P1 FEM: ASM vs sASM (CG iterations)



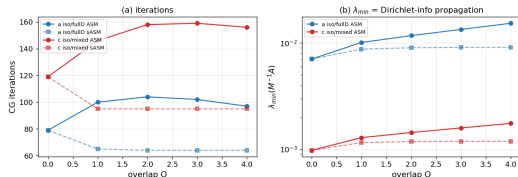
3D — solid ASM, dashed sASM

Analysis. ASM: iters rise with overlap (a 79→104, c 119→158, **worst d-3D 203→413**); sASM: monotone down (d-3D 203→143). 1D: ASM flat (ICC exact, $\omega=1$), sASM slightly worse — no over-count to remove.

Conclusion. sASM beats ASM across all 2D/3D cases; hurts only in 1D. Worst = 3D + mixed BC + anisotropy.

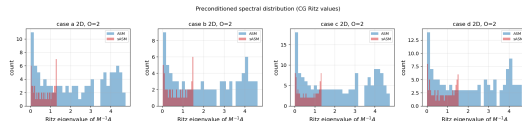
Result: condition number, dimension, boundary, anisotropy

BC effect: full-Dirichlet (a) vs Dirichlet+Neumann (c), 2D



BC effect (a vs c): full-Dirichlet has larger $\lambda_{\min}$ (info from all sides)

Analysis. κ tracks iterations; ω rises to 2–5 on unstructured 3D P1 (ICC weaker on irregular meshes); $\hat{N}$ up to 20 (box corners). sASM tightens the spectrum, lower total time.



Ritz spectrum: ASM cluster of large eigenvalues (over-count) pulled back by sASM

Conclusion. κ , spectrum and iterations agree; the anomaly is the geometric over-count $\hat{N}$ amplified by inexactness.

Does sASM also lower $\lambda_{\min}$? (both spectral ends)

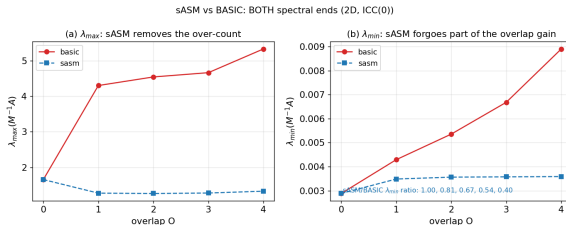
Studies: the Ritz slide showed the distribution; here we track BOTH ends of $M^{-1}A$ for BASIC vs sASM.

Size/params: 2D $N=1089$, $S=4$, $O=0-4$, ICC(0).

Result ($O=4$): $\lambda_{\max}$ $5.33 \rightarrow 1.33$ ($\sim 4\times$ down); $\lambda_{\min}$ $0.0089 \rightarrow 0.0036$ (ratio 0.40).

Net κ : BASIC 599 $\rightarrow$ sASM 369 — still better.

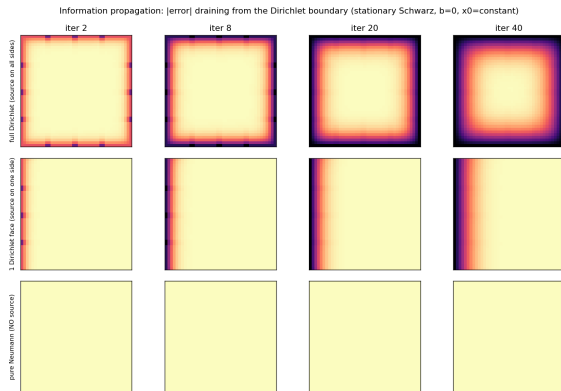
Analysis. The $D^{-1/2}$ weighting scales down the overlap region, so sASM *forges part of the overlap* $\lambda_{\min}$ *gain* (ratio 1.00 $\rightarrow$ 0.40) — yes, $\lambda_{\min}$ *drops*. But $\lambda_{\max}$ falls $\sim 4\times$ vs $\lambda_{\min}$ only $\sim 2.5\times$, so net κ improves.



sASM cuts $\lambda_{\max}$ (over-count) hard; it also lowers $\lambda_{\min}$, but less.

Conclusion. sASM lowers *both* ends; the large $\lambda_{\max}$ reduction dominates the smaller $\lambda_{\min}$ loss $\Rightarrow$ net κ down. The remaining small $\lambda_{\min}$ is the coarse-space target (Part III).

How Dirichlet data propagates: with vs without a source



Setup: stationary (damped) Schwarz, $b=0$, $x_0=\text{constant}$ (the global mode); plot |error| per sweep.

Read it as: the Dirichlet boundary is the *source* — error drains *from* it inward, $\sim$ one subdomain/sweep.

full Dirichlet: drains from all 4 sides $\Rightarrow$ fast, large $\lambda_{\min}$.

1 Dirichlet face: drains from one side only; far side lags $\Rightarrow$ small $\lambda_{\min}$.

pure Neumann: flat at every iter — *no source*, the constant can't drain ($\lambda_{\min} \rightarrow 0$, singular = Sys2).

Conclusion. “Information propagation” = the boundary value draining inward each sweep. **More Dirichlet boundary = more sources = faster fill = larger $\lambda_{\min}$.** Pure-Neumann has no source $\Rightarrow$ the global mode is undamped $\Rightarrow$ Sys2 is hardest — exactly what a coarse space must supply.

Part II

Attribution

which factor makes it hard, via which channel

PART II

Attribution: dimension $\rightarrow \lambda_{\max}$, **boundary/shift** $\rightarrow \lambda_{\min}$

Studies: is it 3D, Dirichlet, Neumann, or shift that makes it hard?

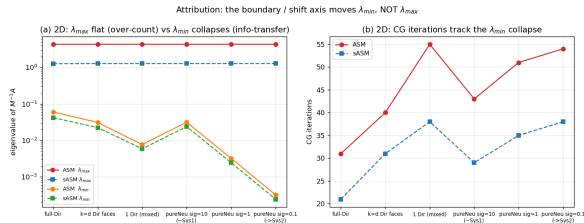
Problem: $A = K + \sigma I$ (σ large $\approx$ Sys1, $\sigma \rightarrow 0 \approx$ Sys2), FD grid.

BC axis: full / k -Dir faces / 1-Dir (mixed) / pure-Neumann $\sigma \in \{10, 1, 0.1\}$.

Size: 2D $M=25$ ($N=625$), 3D $M=13$ ($N=2197$); $S=4$.

Params: overlap 0–3; ICC(0); ASM & sASM.

Result: $\lambda_{\max}=2D \ 4.30/3D \ 8.31 (= \hat{N})$, FLAT over BC/ σ ; $\lambda_{\min}$ collapses $\sim 190 \times (2D)/300 \times (3D)$; $\kappa \propto 1/\sigma$.



$\lambda_{\max}$ flat (over-count, set by dimension); $\lambda_{\min}$ collapses (info transfer, set by boundary/ σ); iters track $\lambda_{\min}$.

Conclusion. No single culprit — **dimension sets $\lambda_{\max}$ (over-count $\hat{N}=2^d$)**, **boundary/shift set $\lambda_{\min}$** ; they are orthogonal and multiply. Sys2 ($\sigma \rightarrow 0$, pure-Neumann, singular) = hardest. Overlap/sASM cannot touch $\lambda_{\min} \Rightarrow$ need a coarse space.

Part III

A non-GenEO coarse space

shift-invariant spectral + known-coefficient modes

III.a Shift-invariant (K, M) spectral coarse

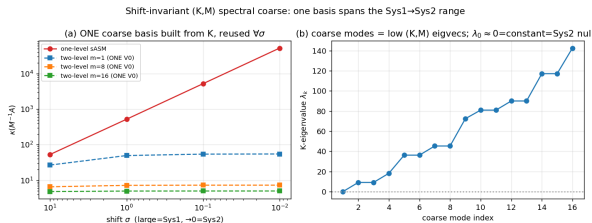
Studies: one coarse basis valid across the whole $\text{Sys1} \rightarrow \text{Sys2}$ (σ) range.

Problem: pure-Neumann K , $A = K + \sigma I$.

Size: 2D $M=25$ ($N=625$), $S=4$, $O=1$.

Params: $\sigma \in \{10, 1, 0.1, 0.01\}$; coarse modes $m \in \{1, 4, 8, 16\}$; V_0 = low eigvecs of K , built ONCE.

Result: one-level κ 53 $\rightarrow$ 52000 as $\sigma \rightarrow 0$; two-level $m=1$ (constant) caps $\kappa \approx 54$ flat; $m=16 \Rightarrow \kappa \approx 5$.



$A_0 = \Lambda_{low} + \sigma I$: same V_0 for all σ ; $\lambda_0 \approx 0$ mode = constant = Sys2 nullspace .

Conclusion. One basis (built once from K) spans $\text{Sys1} \rightarrow \text{Sys2}$; the constant mode deflates the Sys2 singularity ($m=1$ already removes the $\sigma \rightarrow 0$ blow-up).

III.b Known-coefficient A-harmonic modes (non-GenEO)

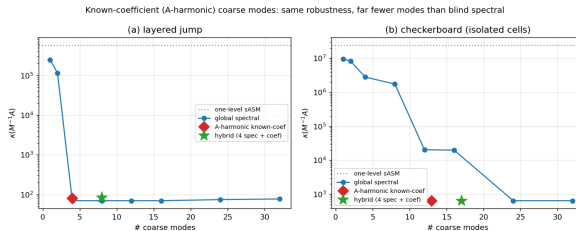
Studies: handle high-contrast without GenEO eigenproblems.

Problem: $-\nabla \cdot (a \nabla u) = f$, mixed BC, FD.

Size: 2D $M=25$ ($N=625$), $S=4$, $O=1$.

Params: $\text{coef} \in \{\text{const, layers, checker}\}$, $\rho=10^4$;
constructions = global spectral / raw indicator /
A-harmonic / hybrid.

Result: checker: A-harmonic 13 modes $\Rightarrow \kappa=646$ (= global spectral $m=32$); hybrid (4 spec+coef) $30\times$ better than spectral-only at equal modes.



A-harmonic known-coef modes reach the robustness floor with $\sim$ half the modes of blind global spectral.

Conclusion. Build contrast modes from the **known** conductivity regions (cardiac fibers/ σ_i , σ_e) instead of discovering them — same robustness, far fewer modes, no eigenproblems.

Part IV

Cross-system reuse

Sys1→Sys2 (same mesh), Sys3 (different mesh)

IV.a Sys1 → Sys2: one build-once shared coarse

Studies: reuse one hybrid coarse across Sys1 → Sys2 + a timestep sequence; beat Task-3's $1.6\times$.

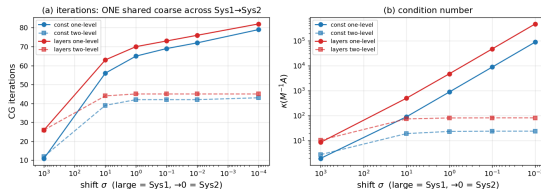
Problem/BC: shared heart mesh, $A = K + \sigma I$, pure-Neumann.

Size: 2D $M=33$ ($N=1089$), $S=4$, $O=1$.

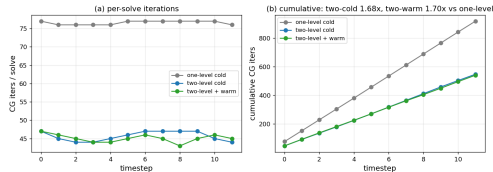
Params: $\sigma \in \{10^3..10^{-4}\}$; coef const/layers($\rho=10$); hybrid coarse (4 spectral + A-harmonic); sequence $T=12$ moving front at $\sigma=10^{-2}$.

Result: κ flat over σ (one-level $\propto 1/\sigma \rightarrow 4.7e5$); sequence **919**→**548** (**1.68** $\times$), +warm 541 (**1.70** $\times$).

Cross-system shared coarse: built ONCE, flattens the whole Sys1→Sys2 range



Sys2-like timestep sequence: shared coarse + warm start (analog of Task-3)



Conclusion. One coarse, built once, flattens the whole Sys1→Sys2 range; **coarse is the main lever (1.68 $\times$)**, warm-start **marginal** — matches Task-3.

IV.b Sys3 (torso): cross-mesh interface scheme

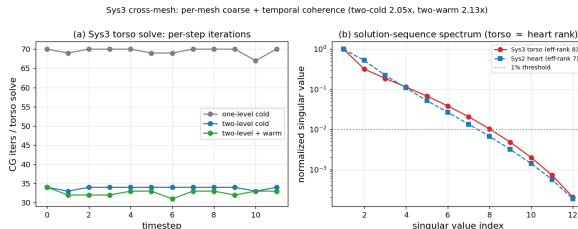
Studies: a different mesh, coupled only by the epicardial interface — reuse via per-mesh coarse + temporal coherence.

Problem/BC: torso Laplace; $x_0=0$ face = interface Dirichlet (u_e , moving front), rest Neumann; *non-singular*.

Size: torso $M=41$ ($N=1681$) $\neq$ heart $M=33$; $S=4$, $O=1$.

Params: sequence $T=12$ moving-front interface data; per-mesh coarse $m=8$; one cold / two cold / two warm.

Result: 834 $\rightarrow$ 406 ($2.05\times$), +warm 391 ($2.13\times$); eff-rank 8 $\approx$ Sys2's 7.



Per-mesh coarse is the lever ($2.05\times$); warm-start marginal — the moving front is moderate-rank on either mesh.

Conclusion. Cannot share the heart V_0 , but the *same construction* on the torso mesh gives $2.05\times$. **Honest negative:** “torso is much more warm-startable” is *not* supported (eff-rank 8 $\approx$ 7).

Unifying conclusions

- 1 **The anomaly is real and mechanistic:** unscaled BASIC ASM + inexact ICC makes overlap raise $\lambda_{\max}=\omega\hat{N}$; sASM removes $\hat{N}$ and wins in 2D/3D (hurts only in 1D where $\omega=1$).
- 2 **Two orthogonal channels:** dimension sets $\lambda_{\max}$ ($\hat{N}=m_{\text{axis}}^d$); boundary & shift set $\lambda_{\min}$. Sys2 (pure-Neumann, $\sigma\rightarrow 0$) is hardest because $\lambda_{\min}\rightarrow 0$ (singular).
- 3 **A non-GenEO coarse fixes $\lambda_{\min}$:** shift-invariant (K, M) spectral modes (smooth + constant/nullspace, reused across σ) + known-coefficient A-harmonic modes (contrast, few modes, no eigenproblems).
- 4 **Coarse is the cross-system lever:** Sys1 $\rightarrow$ Sys2 (same mesh) $1.68\times$, Sys3 (different mesh) $2.05\times$; warm-start/POD marginal everywhere (moving front is moderate-rank) — reproducing Task-3 across all three systems.

Limitations & next steps

Limitations (honest)

- Coarse-space results on a NumPy exact-spectrum proxy with scalar-contrast fields, not the real tensor anisotropy.
- Speedups ($1.68\times$, $2.05\times$) are proxy values benchmarked against Task-3's real-pipeline $1.6\times$.
- $\sigma=0$ exact singular handled by nullspace deflation (constant in V_0).

Next steps

- Port the A-harmonic construction to the FEM fiber/conductivity tensor (cardiac σ_i, σ_e).
- Scale up in PETSc 3.24 on the real cardiac mesh; re-author `xsys_precond.c`.
- Front-following interface enrichment for Sys3 (minor refinement per the data).

Conclusion. Tooling: `asm_spectral.py`, `schwarz_fem.c`, `coarse_proto.py`, `xsys_proto.py`, `sys3_proto.py`; all data & figures on branch `influencing-factors-discussion`.